



Developing and Validating the Artificial Intelligence Literacy Concept Inventory: an Instrument to Assess Artificial Intelligence Literacy among Middle School Students

Helen Zhang¹ · Anthony Perry² · Irene Lee³

Accepted: 17 February 2024 / Published online: 5 May 2024
© International Artificial Intelligence in Education Society 2024

Abstract

The rapid expansion of Artificial Intelligence (AI) in our society makes it urgent and necessary to develop young students' AI literacy so that they can become informed citizens and critical consumers of AI technology. Over the past decade many efforts have focused on developing curricular materials that make AI concepts accessible and engaging to young learners; and yet, limited research investigated how to assess learners' AI literacy, which is critically important to inform the teaching and learning of AI. This paper addresses this issue by reporting the development and validation findings of the AI Literacy Concept Inventory Assessment (AI-CI), a set of multiple-choice questions designed to assess understanding of AI literacy concepts among middle school students. The AI-CI consists of 20 multiple choice questions examining student understanding of four topics: AI general concepts, logic systems, machine learning general concepts, and supervised learning. The content validity of AI-CI was established through multiple rounds of expert panel reviews with AI educators and experts, observations of student learning of AI, and cognitive validation interviews. The validity of the AI-CI was established with a sample of 981 students and the pre-posttest reliability was established with a sample of 108 middle school students who learned AI through experiencing the Developing AI literacy (DAILY) curriculum. The findings show that the AI-CI is a valid and reliable tool to assess AI literacy at the middle school level.

Keywords AI literacy education · Middle school · Assessment

✉ Helen Zhang
zhangzm@bc.edu

¹ Lynch School of Education and Human Development, Boston College, Chestnut Hill, MA, USA

² College of Education and Human Development, University of North Dakota, 231 Centennial Drive Stop 7189, Grand Forks, ND, USA

³ Scheller Teacher Education Program, Massachusetts Institute of Technology, Cambridge, MA, USA

Introduction

With the rapid growth in the availability of enormous amounts of data and computation power, Artificial Intelligence (AI) is becoming widely used across fields, including manufacturing, economy, communications, transportation, medical care, and education. AI's increasing impact on industry and society necessitates developing a future generation of AI literate citizens who can utilize their knowledge of AI to make civic decisions and workforce who are comfortable with working with AI tools and technologies (Cantú-Ortiz et al., 2020; National Science Board, 2020). Educators have recognized this urgent need and started developing curricula and approaches to foster AI literacy among learners at the K-12 level. Yet many of the current efforts have focused on creating novel activities that are accessible and engaging for young learners (e.g., Ali et al., 2021; Eguchi et al., 2021; Williams & Breazeal, 2020). They often evaluate the impact of educational innovations using qualitative measurements such as engagement and interest. Only a few research studies reported student learning outcomes using quantitative measures of conceptual understanding (Lin et al., 2020; Zhang et al., 2023). This gap illuminates a challenge for the nascent field of AI literacy education—without reliable and valid assessment tools on student conceptual learning, how can we know which AI literacy concepts students are capable of learning? How can we determine the effectiveness of various interventions on helping students build understanding of AI literacy concepts?

Our study helps address this concern by reporting the development and validation of the AI Literacy Concept Inventory assessment (AI-CI), an assessment tool that aims to measure students' conceptual understanding of AI literacy at the middle school level. We define AI Literacy as the subset of AI knowledge and competency relevant to becoming an informed citizen and critical consumer of AI. In particular, the AI-CI was developed purposefully to measure the AI literacy of youth from groups underrepresented in STEM/Computing for multiple reasons. First, we seek to broaden the participation of students from underrepresented groups in STEM/Computing and thus it is essential to develop and validate instruments that are relevant and interpretable to youth from these groups. With these tools we can more accurately measure their AI literacy and improve the curriculum design to better engage youth from underrepresented groups. Second, most youths nowadays may have been exposed to AI through social media tools such as YouTube, Instagram, and Snapchat, and they, particularly youth from historically underrepresented groups, may be targeted by fake media or negatively impacted by bias in AI applications. Therefore, fostering their understanding of machine learning techniques used in social media curation and generation is imperative for youth to become informed citizens and critical consumers of AI.

The development of the AI-CI entailed quantitative and qualitative approaches: the qualitative approaches included interactive cycles of expert panels, open-ended interviews with students, observations of student learning of AI activities, and cognitive validation interviews to identify student misconceptions and to develop the items; the quantitative approaches included a large scale validation

study with 981 students to investigate external validity using item response theory (IRT) and a reliability study with a sample of 108 middle students before and after they learned with an AI literacy curriculum called the Developing AI Literacy (DAILy) curriculum to test pre-posttest reliability.

Theoretical Background

AI Literacy

With the wide spread of AI in our society and daily lives, more and more researchers and educators realize that literacy in AI and computer science is and will be becoming as important as classic literacy (reading and writing) (Kandlhofer et al., 2016). Thus, the term “AI literacy” has emerged and many researchers tried to provide a definition of the term. For instance, Long and Magerko (2020) through a synthesis of the published work provided a concrete definition of AI literacy that included a comprehensive set of competencies “*that enables individuals to critically evaluate AI technologies, effectively communicate and collaborate with AI, and utilize AI as an online tool in both homes and workplaces*” (p.2). They further provided a conceptual framework of AI literacy composed of five different overarching themes derived from the literature including “What is AI?”, “What can AI do?”, “How does AI work?”, “How should AI be used?” and “How do people perceive AI?.” Ng et al. (2021) proposed four aspects of AI literacy based on their exploratory review of existing peer-reviewed articles: knowing and understanding AI (knowing the basic functions of AI and how to use AI applications), using and applying AI (applying AI knowledge, concepts, and applications in different scenarios), evaluating and creating AI (higher-order thinking skills with AI applications), and AI ethics (human-centered considerations). They suggested to add AI literacy as an essential component of the twenty-first century digital literacy, a fundamental skill that everyone needs to possess in order to thrive in work settings and everyday life, not just computer scientists.

Although researchers have not reached a consensus regarding the definition of AI literacy, they all agreed that AI literacy overlaps with other disciplines such as data literacy or computational literacy. For instance, Ng et al.’s review found eight papers borrowed the ideas of Brennan and Resnick (2012)’s framework of computational thinking to conceptualize AI literacy. These papers attempted to define AI literacy through the lenses of concepts, practices, and perspectives. In a recent study (Celik, 2023), computational thinking was found to be positively associated with AI literacy, i.e., individual’s computational thinking skills could facilitate their use, recognition, and evaluation of AI-based technologies. Meanwhile AI literacy is distinct from other disciplines in that it incorporates data science and computational thinking into a multidisciplinary concept and highlights ethical concerns.

In the field of K-12 AI education, some researchers have explored AI literacy for K-12 students because young learners typically have limited mathematical background prerequisite for AI courses at the post-secondary or professional levels and limited cognitive capacities for processing complex concepts. Thus, AI literacy

for K-12 learners is different from AI literacy for adults. The AI4K12 initiative (Touretzky et al., 2019), through their workgroup with educators, AI experts, and school administrators, delineated five big ideas that are sufficient to cover the richness of the field of AI but small enough to be manageable by K-12 teachers and students, including (1) perception (i.e., computers perceive the world through sensing); (2) representation and reasoning (i.e., agents maintain representations of the world and use them for reasoning); (3) learning (i.e., computers can learn from data); (4) natural interaction (i.e., intelligent agents require many types of knowledge to interact naturally with humans); and (5) societal impact (i.e., artificial intelligence can impact society in both positive and negative ways). Other educators have developed learning resources to explore how to teach AI literacy concepts that are critical for students at the K-12 level, such as Universidade da Coruña's AI+ Curriculum for European High School (Guerreiro-Santalla et al., 2020), MachineLearningforKids in the UK (Lane, 2021), and AI4K12 initiative's K-12 Guidelines for Artificial Intelligence (Touretzky et al., 2019). These learning resources often used engaging, hands-on, and non-mathematical activities to teach AI and the associated research studies focused on student engagement with the learning activities. As such the field of K-12 AI education remains inconclusive of what AI literacy means at the K-12 level, what concepts and skills AI literacy at the K-12 level entails, and what AI concepts are age appropriate for young learners.

Challenges and Opportunities for an Established Assessment for AI Literacy Education

Assessment plays a critical role in informing and improving learning and teaching. Through appropriate assessment tools, teachers can classify and grade their students to evaluate their learning acquisition, give specific feedback to students in support of their learning, and structure their teaching accordingly (Cowie & Bell, 1999; Taras, 2005). This is particularly critical to the nascent field of K-12 AI education given the blossom of AI focused educational interventions in K-12 classrooms and afterschool settings. While there exist assessments associated with standalone courses or curricular innovations on AI, these instruments are often developed to assess localized contextual and conceptual factors of the curricula (e.g., machine learning, gamified design) or to examine student learning outcomes in the affective domains (e.g., interest, self-efficacy). There is a significant need for a more general AI literacy assessment to examine student conceptual learning for specific age groups (e.g., middle school or elementary school grade bands). Such established assessment mechanisms could provide researchers and K-12 educators with an anchor with which to validate the effectiveness of new curricular and/or pedagogical innovations. This will ultimately speed up the pace of broadening participation in AI education as educators would have objective judges of the impact on students and no longer struggle with contradictory results from various evaluation mechanisms.

Yet developing a general assessment to measure conceptual learning is not easy for the young field of AI literacy. Some of the difficulties in developing a general AI literacy assessment are shared with other STEM and Computer Science disciplines

while others are unique to the AI education field. For instance, one of the biggest challenges is to determine the breadth and depth of AI literacy concepts covered in the assessment. This is particularly difficult given the lack of a clear definition of AI literacy at the K-12 level and a dearth of understanding about how deeply young students can learn of AI. Further, similar as the Computer Science and other technology fields, the content covered in AI literacy inevitably changes with the changes in the technologies we use. This is especially true given the fast development of the AI technologies. This poses a challenge to the assessment development because the items need to be technology-independent while easily adaptable to withstand present and future changes at the same time. Another challenge is that fostering AI literacy includes developing not only an understanding of AI concepts but also the processes (e.g., identifying and mitigating potential bias of an AI model, modifying, testing AI models). Similar as processes of debugging and testing in Computer Science (Taylor et al., 2014), this aspect of learning is difficult to acquire but important to assess because it constitutes one of the central practices of real-life work in the field. Students need to first understand the underlying concepts and then apply their understanding to the process to solve problems. Assessment developers must navigate the interplay between conceptual understanding and application of that understanding to problem solving situations. Finally, the assessment should be valid, meaningful, and practical so that teachers can easily adopt and administer in their classrooms.

Potentials of a Concept Inventory for AI Literacy Education at the K-12 Level

We posit that a conceptual inventory (CI), an assessment tool that employs “distractor-driven” multiple-choice questions to measure student understanding of the core concepts of a topic (Goldman et al., 2010; Taylor et al., 2014), can potentially meet the evolving needs and benefit the field of AI literacy education. Different from comprehensive assessments that cover everything taught in a course, CIs focus on fundamental concepts of AI literacy, meaning that all students should be capable of performing well on it. This makes CIs easy to administer (Almstrum et al., 2006).

The design of CI items is deeply rooted in qualitative cognitive research and focuses on incorporating attractive distractors that represent misconceptions commonly held by students as answer choices. Hence the premise of a CI is that students have developed prior ideas related to the target concepts. This aligns well with research findings that K-12 youth nowadays have developed naïve ideas about AI through their prior experiences. For instance, many students have been interacting with AI-enabled technologies in home such as Siri and Google Home on a daily basis. They may develop emotional connections with AI technologies (Druga et al., 2017; Turkle et al., 2006) and believe that robots are smarter than human (Williams et al., 2019). They also may have developed some understandings of how to work with AI-enabled tools through using technologies that integrate generative modeling techniques such as photo filters embedded in social media apps. Many youth at the elementary and secondary school level may also have developed ideas regarding the negative aspects of AI and tended to describe more dangers associated with AI than its benefits and potentials when talking about AI (Kreinsen & Schulz, 2021). All

these findings suggest that it is critical that AI literacy education address and correct student misconceptions and that an assessment of AI literacy detect student misconceptions and provide accurate, detailed information of students' knowledge of AI. CI's focus on creating distractors based on students' commonly held misconceptions makes it capable of probing into students' potential mismodeling of concepts. The distractors chosen by test-takers, in turn, can help researchers understand student thinking and provide insights into students' prior knowledge.

Further, compared to the traditional standardized assessments that are generally contracted to testing companies that rarely employ the research literature on student misconceptions as part of their item development, CIs have shown greater effectiveness in detecting conceptual understanding of students. Many research studies have revealed that using CIs enables teachers to monitor student learning and identify what misconceptions are stubborn among students (Costa et al., 2008; Sadler et al., 2010; Savinainen & Scott, 2002) and that the distractors provide greater diagnostic utility than typically multiple-choice items, while retaining their efficiency advantages (Briggs et al., 2006). Meanwhile researchers in psychometrics also found that the distractor-driven items are more difficult than traditional standardized test items because of the inclusion of popular misconceptions as the distractors (Ascalon et al., 2007; Morris et al., 2006). Yet we argue that this inclusion of distractor-driven items well serves the purpose of assessing student conceptual understanding. If the most prominent misconceptions are not included as distractors in test items, such items do little to inform teachers of their students' initial ideas, particularly when administered as a pretest prior to instruction. Neither do such items sufficiently measure how solid students' newly developed ideas are, given that students are not exposed to attractive misconceptions in the answer choices and students may choose an answer based on what a teacher said in class. Given the nascent state of the field AI education at the K-12 level, it is critical that the assessment does not mislead educators into believing that students have mastered a particular concept because they choose the correct answer. Hence items that employ distractors based on common misconceptions are of great potentials for informing the teaching and learning of the field.

In addition, the large impact of the Force Concept Inventory (FCI) on physics education (Hestenes et al., 1992) can best exemplify how concept inventory assessments can accelerate a field and revolutionizing teaching and learning. The FCI was first developed to test students' conceptions of force and motion as covered in their first college physics course. Administering the FCI revealed a vast difference between what teachers thought their students learned based on traditional final exams and the misconceptions their students actually held (Adams & Wieman, 2011; Mazur, 2009). This finding sparked widespread change in physics education, led physics educators to rethink their teaching practices, and inspired innovative curriculum design and pedagogies such as Peer Instruction and other active learning interventions that resulted in significantly improved conceptual learning outcomes (Crouch & Mazur, 2001). Researchers in other STEM disciplines have ever since successfully developed and used CIs in various domains, including astronomy and space science (Sadler et al., 2010), chemistry (Krause et al., 2004), geoscience (Libarkin et al., 2006), and biology (Garvin-Doxas & Klymkowsky, 2008). In the field of computational thinking and Computer Science, similar assessment tools

have been developed such as the Middle Grades Computer Science Concept Inventory (MG-CSCI) Assessment (Rachmatullah et al., 2020) and the Concept Inventory for Introductory Programming (Caceffo et al., 2016). By focusing on a set of central concepts and incorporating students' misconceptions in the design of the items, CIs have proven to be a valuable tool to accurately assess student domain-specific learning and evaluate the effectiveness of curricular and pedagogical innovations across domains.

Developing the AI Literacy Concept Inventory Assessment (AI-CI)

Motivated by CIs in STEM and considering the field's need, we consider an CI for AI education necessary and imperative. Our ultimate goal is to develop a validated CI that is independent of any specific curricula and well aligned with an emerging set of AI concepts and processes that are age-appropriate for middle schoolers. Given the lack of a consensus of what AI literacy for middle schoolers means, we defined AI literacy at the middle school level as the subset of AI knowledge and competency relevant to becoming an informed citizen and critical consumer of AI. To make the definition operational for the development of concept inventory (i.e., identifying a set of age appropriate concepts that could be assessed with multiple-choice items), we purposefully developed the AI-CI concurrently with the Developing the AI Literacy (DAILY) curriculum, which consists of a set of AI curricular activities for young learners based on prior work (Ali et al., 2019). We started the development of the assessment and the curriculum with a literature review and expert panel reviews to identify a set of AI concepts that are critical, fundamental, and age appropriate to middle school students (see Section "[Establishing the Topics and Content Validity via Expert Panels \(Step 1\)](#)" and Table 1). Once we have a draft of the test, we administered it to a group of students before and after they learned with the DAILY curriculum to ensure that the test is reliable and sensitive to curricular interventions of AI (see Section "[Quantitative Test for Reliability \(Step 6\)](#)"). By doing this, we expect that the AI literacy concepts taught in DAILY that have proven to be age appropriate will serve as the emerging target concepts assessed in the AI-CI.

The development of the AI-CI was informed by Adams and Wieman (2011)'s work on a systematic procedure for building and validating a concept inventory, which has been utilized in the development of multiple concept inventories in computer science at the undergraduate level (Caceffo et al., 2016; Taylor et al., 2020). Considering that AI education is new to most teachers and students may not have much prior knowledge of AI, we modified Adams and Wieman (2011)'s procedure when developing the AI-CI questions. Our process included six steps:

- Step 1. Establish core topics and concepts that are important to teachers and age-appropriate for students via expert panel;
- Step 2. Identify and probe student thinking about these concepts and the various ways it can deviate from expert thinking through observation of student reac-

Table 1 Target topics assessed in the AI literacy questionnaire

Topics	Concepts	Competency
AI general	Intelligence, dataset, learning algorithm,	Recognizing AI: Distinguishing between technological artifacts that use and do not use AI; Recognizing technological tools used in everyday life that may use AI Understanding intelligence: Analyzing features that make an entity “intelligent”, including identifying whether a technology utilizes datasets and algorithms
Logic systems	Decision tree, classification, human logic	Understanding knowledge representation: Understanding what a decision tree is Building decision trees: Understanding how to construct decision trees; Understanding that training and testing a decision tree based on key features in the dataset Understanding decision-making: Recognizing and describe how computers reason and make decisions based on decision trees; Distinguishing the difference between human decision-making and computer decision-making
Machine learning general	Difference between supervised and unsupervised learning, <i>*AI generating vs. classifying</i>	<i>*Recognizing generative AI: Distinguishing between AI that classifies and generates</i> Recognizing supervised learning: Distinguishing between supervised and unsupervised learning (supervised learning utilizes labeled data whereas unsupervised learning utilizes unlabeled data)

Table 1 (continued)

Topics	Concepts	Competency
Supervised learning	Labeled dataset, process of ML (training phase, testing phase), <i>*bias in datasets and algorithms, **how neural networks learn (back propagation, feed forward, evaluation)</i>	<i>Understanding ML processes:</i> Understanding the steps involved in machine learning (how the learning takes place); <i>**understanding the processes of neural networks</i> Understanding AI learns from data: Understanding that ML models learn from data (how the training examples provided in an initial dataset can affect the results of a trained ML model) Recognizing potential bias of AI: Recognizing potential bias issues of supervised learning; Recognizing that humans play an important role in programming and choosing datasets to train AI models; Identifying potential sources of bias in AI
<i>**Unsupervised learning</i>	<i>**Generative Adversarial networks (GANs), How GANs work, generator, discriminator</i>	<i>**Understanding GANs fundamentals: Understanding that the generators and discriminators work against each other in GANs</i>

* Concepts added based on the suggestion of expert panelists

** Concepts removed based on the suggestions of expert panelists

tions to sample DAILY activities and open-ended interview questions with sample students;

- Step 3. Create a forced answer test that measures student thinking and validating with AI experts;
- Step 4. Carry out validation interviews with sample students on the test questions;
- Step 5. Administer the assessment to students, run statistical tests on the results (validity, difficulty level, etc.), and modify items as necessary.
- Step 6. Administer the assessment to students before and after learning the DAILY curriculum, conduct statistic tests on reliability, and change items if necessary.

Key elements of the CI development were the generation of distractors rooted in student misconceptions as revealed in student observations and interviews carried out in Steps 2, 3, and 4. The first two steps focused on establishing content validity of the AI-CI (whether it assesses AI literacy concepts that are important and age appropriate for middle school youth). Afterwards, instead of creating open-ended survey questions to probe student thinking in test form as in Adams and Wieman's procedure, we created a pool of test items (Step 3) and examined the cognitive process validity of the items (whether the cognitive processes students used to answer the questions mirror those intended by the developers) through multiple rounds of cognitive interviews in Step 4. The main reason for our modifications of Adams and Wieman's procedure in Steps 3 and 4 was that the field of AI literacy at K-12 level is new and students are increasingly exposed to AI. Cognitive interviews can reveal more detailed information regarding student conceptions of AI and how they think in test form than an open-ended survey. Further we also modified Adams and Wieman's procedure in that we employed a large validation study in Steps 5 to examine the validity of the assessment (i.e., does the test accurately measure what it is supposed to measure, how difficult). The validation was also used to reduce the number of items so that teachers can administer it in classrooms, ideally within 30 min as requested by middle school teachers. In Step 6 we focused on the reliability of the assessment (i.e., how consistent and stable the test is in terms of measuring what it is supposed to measure, does the assessment sensitive to educational interventions) so that the test can be used as pre and posttest to determine the effectiveness of instructional or curricular innovations.

Establishing the Topics and Content Validity via Expert Panels (Step 1)

The first step of developing the AI-CI was to make the definition of AI literacy operationalizable by identifying which AI literacy topics are important and age-appropriate for middle school youth. As noted above, the AI-CI was developed to examine student understanding of AI literacy concepts and competencies that are necessary for preparing them to become informed citizens and critical consumers of AI in the future. To better operationalize this definition of AI literacy for assessment development, we first curated a list of AI concepts and competencies that align with the definition, have been suggested to be age-appropriate based on our previous work, and literature on effective AI curricula for younger learners (Ali et al., 2019; Williams

et al., 2023; Zhang et al., 2023) and literature reviews of AI literacy competence (Long & Magerko, 2020; Zhou et al., 2020). An initial list of the AI literacy topics included: AI general, logic systems, machine learning general, supervised learning, and general adversarial networks (GANs). See Table 1 for the list. Each topic included a set of core concepts that we think middle school students are capable of learning and competencies that they are able to develop. For instance, the topic of “AI general” at the middle school level entails correct understandings of concepts of intelligence, dataset, and learning algorithm. The two associated competencies include (1) recognizing of AI (being able to distinguish between technologies that use and do not use AI); and (2) understanding intelligence (being able to analyze features of a technology that make it intelligent). The initial list included GANs as a topic because of the widespread of AI-generated media with which students may interact on a daily basis.

Next, we conducted two rounds of expert panels to review and revise this set of core AI topics and concepts. Following suggestions from Taylor et al. (2020)’s work on developing a CS concept inventory, we purposefully selected the panelists who regularly attend the same annual conferences as our team (e.g., Computer Science Teacher Association (CSTA), Association for the Advancement of Artificial Intelligence’s Symposium on Educational Advances in AI (EAAI), Association for Computing Machinery’s Special Interest Group on Computer Science Education (SIGCSE)) and are well-situated to discuss AI education topics. The first panel consisted of seven experienced CS/technology teachers or educators. Considering that not all the CS educators have sufficient background knowledge of AI, we developed a suite of materials for our panelists to review. The materials included our definition of AI literacy, a short description of each topic, and an outline of learning activities we plan to develop to teach the topic to middle school students. We sent out these materials to the panelists with explicit questions asking for their feedback and suggestions on the age appropriateness and importance of these topics to middle school aged youth and whether the list needs to include other relevant topics. The second panel included four AI content experts (i.e., computer scientists from universities and the industry), who were asked to comment not only the age appropriateness of the topics but also whether these concepts and the proposed curricular outline were accurate and important to the field of AI education.

Overall, the panelists felt these topics were important for developing AI literacy among middle schoolers and provided rich feedback on the age-appropriateness of the concepts. Below list major suggestions and feedback from the panelists (each panelist was given a unique ID for the research purpose):

- Focus on four topics as an initial set of core concepts to be assessed in AI-CI: AI general, logic systems, ML general, and supervised learning. Panelists expressed concerns about whether topics Neural networks and GANs (feed forward, back propagation, etc.) were age appropriate for middle school students. For instance, panelist DT (an AI expert) expressed that *“Given lots of experience with middle school CS/CT but not AI and given a little bit of experience with AI for upper elementary, my guess would be it’s too complex for upper elementary and middle school.”* Two panelists explained similar ideas and thought topics of Neural net-

works and GANs are appropriate for 8th graders but not early middle school grades (6/7th grade), i.e., *“I’m worried about the 11 year olds... My sense is that the older group is more ready to take on these abstract ideas and be ready to thoughtfully engage in these ideas than the younger group. (panelist BL)”* and *“I wondered if the ideas about back propagation and gradient decent will be a stretch. These are the more difficult topics. I think you will need to experiment with the younger age group. (panelist CB)”*

- Add a topic of AI related ethics issues. Two educator panelists suggested emphasizing the ethics framework in the curriculum and assessing students’ understanding of bias issues related to AI, e.g., *“Kids at this age are impacted consciously and unconsciously by AI... It is good for their brains to try to understand positive and negative aspects of issues—including AI... Letting kids know how AI happens—that it is not magic, seems important. They should be a part of a generation that can use AI rather than be used by it. Understanding how AI is/can be biased—just like other aspects of their world are—and understanding strategies to counter this, may help them create a more just society. This is more important from my perspective. (panelist SK)”*
- Avoid the use of jargon and simplify the language. Three educator panelists expressed concerns about new vocabulary used in the learning activities and assessment items, e.g., *“If you can limit the amount of new vocabulary a bit, please do so... You should use as many analogies and real life examples [in the assessment] because this is abstract for kids. (panelist AW)”*
- Need to assess student awareness of other types of AI (generating vs. classifying). One AI expert (panelist NW) expressed that it is important to expose students to various types of AI, *“AI is more than just machine learning. It is important for students to learn about this. This can probably be added in the AI general [topic].”*

Based on the panelists’ feedback, we finalized a set of core AI literacy concepts that are critical and age appropriate at the middle school level: AI general concepts, logic systems, machine learning, and supervised learning (including ethics). We decided to prioritize the development of items assessing these concepts. Further, to make AI more concrete to students and minimize the use of jargon, AI-CI items would employ scenarios drawn from everyday examples. In questions that have to include jargon, the question statements would provide explanations of the terminology in layman language. We will experiment whether middle school students can comprehend the concepts of neural networks and GANs in the DAILY curriculum: in the DAILY curriculum we specified that AI makes computers perform “intelligent” tasks such as recognizing things, understanding them, and making decisions about them. The curriculum engaged students in learning fundamentals of various AI models, including logic system, machine learning, neural networks, and GANs.

Identifying Student Thinking (Step 2)

The purpose of the second step is to find out students’ naive ideas that can serve as a basis for the “distractor” choices. We first conducted observations of student

learning during a pilot test of three DAILY lessons in an afterschool program among middle school aged youth from groups that are underrepresented in STEM. The lessons included activities introducing students what AI is, how logic systems are built and used, bias of existing AI technologies. Two researchers were present during the pilot test and took observation notes on student reactions (e.g., what questions they asked, what kind of ideas expressed by students, and what they struggled to understand). The observations revealed initial ideas around how students made sense of AI, what prior ideas students brought to the learning, how and in what context they connected with prior ideas. After the observation, we interviewed five students (four were Hispanics/Latino, one was Black) selected to probe more deeply their ideas that may have gone unnoticed during the pilot test. All students for whom we had parent consent and student assent were selected for the study.

Each semi-structured interview lasted approximately 30 min and used guiding questions of “What does artificial intelligence mean to you? Could you tell me an example of AI?”, “If a student wants to train an AI system to tell if a picture has her/him in it. What does she need to do to train the system?”, and “Do you think AI systems may be biased? Why or why not?”. The interview protocol also included questions that probed students’ existing ideas about AI concepts, such as “What do you think the term “artificial neural network” means?”, “Which of the following technology are examples of AI, (a) doors that automatically open when something nears them, (b) a search algorithm of a photo album, (c) a student database of a school district saved in excel that you can search to find a student’s information, (d) a system that randomly predicts weather, and (e) a voice recognition system?”, and “Do you think AI can generate new things?”.

The observations and interviews revealed students’ varied ideas of AI. For instance, four out of the five student interviewees explained that they think “*AI are machines that can do things on their own*” (S3) and “*it needs to be taught so when you ask it, it can talk about it.*” (S5) These answers suggested that students may think that AI can achieve human-level or even higher intelligence. We also found that students had almost no ideas about neural networks or generative AI. None of the student interviewees could articulate any meaningful ideas about these concepts or processes. Even after the interviewers’ explanation of these terms in layman language, students still had difficulties making sense of these concepts. As another example, during the implementation of an activity that asked students to provide examples of technology that they think use AI, students were observed to provide many examples other than robots, such as Alexa, Google home, or Tesla. This suggests that some middle school students have recognized that AI includes more technologies or tools besides robots, which is different from the public’s perception of AI as robot as reported in the literature (Fast & Horvitz, 2017; Ottenbreit-Leftwich et al., 2021; Zimmerman, 2018). One reason for this change may be due to students’ prior exposure to AI-enabled tools in everyday lives like chatbots and image recognition systems.

Further, we found students were confused about some related terminologies when explaining what AI is, including AI, automation, sensing, and algorithms. Three student interviewees thought that as long as the technologies use algorithms, they are AI. For instance, in the question asking them to identify which technologies are

examples of AI, student S1 thought the search algorithm of a photo album, the student database of a school district saved in excel that you can search to find a student's information, and the voice recognition system were AI, *"because they [the searchable student database and the photo album] need to be programmed, in order to search for the photo and the student database. The weather prediction wouldn't be AI because it's random...the voice recognition technology also needs to be programmed to tell what the voice is."* She also thought that automatic doors were AI because *"the door has to be programmed to be able to sense when something is near it, so it must be AI."* Student S4 thought these technologies are all examples of AI because *"they are all digital...I really don't know what algorithms mean but assuming they are all digital, so they all are [AI]."*

We also found students had a hard time learning and interpreting AI related issues. For instance, we observed students' struggles when building an ethical matrix to demonstrate that different stakeholders have different priorities and hence may create biased AI systems. Sixth- and seventh-graders in particular had challenges understanding and imagining who different stakeholders might be and why they may have different perspectives. This may be because young adolescents think in a more egocentric way (O'Connor, 1995).

Creating a Pool of Assessment Items and Conducting Expert Review (Step 3)

Based on the findings, we created a bank of potential questions assessing student understanding of the four topics that panelists deemed age-appropriate. Most of the questions utilized scenario-based questions and incorporated distractors based on student misconceptions identified in Step 2. For each topic, we held a meeting with content experts, assessment researchers, and educators in our team to solicit feedback and refine the questions. The goal of the meetings was to ensure that we as a team felt the questions assessed the target concepts at a difficulty level we felt appropriate and that the scenarios were drawn from everyday events that students were familiar with. Each team meeting prompted revisions to the questions until we reached a consensus that the questions were suitable in the fourth meeting.

Cognitive Validation Interviews with Students (Step 4)

Once we have a pool of questions, the next step was to establish the cognitive validity of the assessment questions, which refers to gathering evidence regarding whether the respondents' cognitive processes during test performance mirror those intended by test designers (Messick, 1995; Yu & Wolters, 2002). A wealth of research has suggested the importance of establishing cognitive validity in assessment development and researchers called for using interviews to gather data around three critical cognitive information processing steps: item interpretation (interpreting item meaning), coherent elaboration (recalling memories relevant to the item), and congruent answer choice (choosing an answer that accurately reflects those memories) (e.g., Woolley et al., 2006). Informed by this body of literature, we developed a cognitive interview protocol to validate the cognitive processing steps students undergo

when taking the forced-answer test. Further given that the AI-CI items employed scenario-based questions, another impetus for the interview was to figure out how students, particularly those from underrepresented groups, made sense of the question statements and the choices and what strategies they used to solve the problems. The interview protocol was developed based on Karabenick et al (2007)'s guidelines of cognitive interviews, the goals of which were to engage students in thinking aloud while solving the problems. We created four major interview questions for each item. We also conducted a short training on how to think aloud before the start of the cognitive interview.

(Question 1) Please read this question. Did you notice anything in the question statement? Was there anything that this question reminded you of?

(Question 2) What is this question trying to find out from you?

(Question 3) Which answer would you choose as the right answer for you? Please talk aloud how you decide to choose the answer.

(Question 4) How confident are you in your selected answer?

In total we conducted cognitive interviews with seven middle school students (three females, four males, two White, one Black, four Hispanic). The sampling strategy (five out of the seven interviewees from the underrepresented groups) was informed by the literature on sampling strategies for interview-based qualitative study (Robinson, 2014) and based on our interest of developing an instrument that can accurately detect AI literacy among youth who are underrepresented in STEM/CS education. The hypothesis was that the findings drawn from the interviews would ensure that the cognitive processes interviewees went through would best mimic those undergone by our target audience.

Each cognitive interview lasted from 30 to 50 min. They revealed student challenges when interpreting the items and indicated suggestions on how to revise and remove the questions. To illustrate how the cognitive interviews were used to revise questions, below we present how we revised one question (item CI19) that aims to assess student understanding of supervised learning (competency assessed: Understanding AI learns from data). Table 2 shows the three versions of the question. The first version of item CI19 (see Table 2 version #1) presents students with a set of labeled images (pictures of swan and duck) that are going to be used to train a supervised learning model and then asks students to determine what the trained AI model would predict based on the labeled dataset. The cognitive interview shows that students had no difficulty selecting the correct choice, even for those who did not have any background knowledge of supervised learning. For example, student S explained his thinking as

“Because when I read it, I look to the pictures, I looked at the words under them. And it says which of the following would you expect the system to predict? And I looked and I said, [choice] A all white birds are swans. No, because there's a white duck. [choice B] All swimming birds are all birds here swimming are swans. [choice C] All birds and dark backgrounds for ducks. No, because the swan on the far right, has the darkest background.

Table 2 The first, second, and third (final) versions of a supervised learning question (CI19)

Version	Question
1	Based on the dataset below, an AI system is tasked with predicting ducks and swans.  DuckSwanDuckSwan Which of the following would you expect the system to predict? A) All white birds are swans B) All swimming birds are swans C) All birds on a dark background are ducks D) All birds facing right are ducks
2	An AI system was trained to predict whether a dog is about to bite or not based on this data set.  BiteBiteNot BiteNot BiteBite Which of the following will the system predict? A. All animals on dark backgrounds might be predicted to bite B. All small dogs might be predicted to bite. C. All the dogs with mouth open will bite. D. Since the labels are incorrect or missing, this system will not make accurate predictions
3	An AI technology has been trained to classify images only using the following data: Banana Broccoli Banana Broccoli  Which of the following would you expect the technology to predict? ☐ All things in multiples are bananas ☐ All yellow things are bananas ☐ All things in multiples are broccoli ☐ All crunchy things are broccoli

And [choice D] all birds facing right are ducks, and one of the swans is not facing right. So the only logical answer is B.”

He further explained that he thought the question was trying to test “if we can predict or what an AI system we think can predict based on identifying features to help give a solution or not. You’re trying to see if we can identify characteristics

that belong to a certain group. So AI can do this...” Student S’s explanation shows that he answered this question by simply checking whether the choices matched with the labeled images. His reasoning did not reveal any information about how a supervised learning model makes predictions (e.g., identifying patterns based on the labeled data). This suggested us to carefully state the choices so that students cannot answer the question by matching with the pictures.

Based on the feedback, we created a second draft of the question (see Table 2 version #2), which included a set of five pictures of dogs and asked students to decide what an AI system trained using the pictures can predict. The cognitive interview with the new question showed that students had no difficulty interpreting the question statement yet they interpreted the images and the choices in various ways. For instance, Student L questioned the pictures based on her experience with dogs,

“[I noticed that] One, the dogs are cute. Two, I think it’s weird because two of the dogs have their mouths open. But one of the dogs you can’t see whether its mouth is open or not but it still says that they’re biting. You have the ones where their mouths open but not bite.”

Another student interviewee (student P) questioned the dark background of the pictures in the choices,

“the first choice that all animals in dark backgrounds might be predicted to bite is not true, because the first one [picture] doesn’t really have a darker background. Actually the last one [picture] doesn’t really have a dark background, too. So I’m not sure...”

Student L also commented on the background of the pictures, *“it’ll be huge to say that they’re [dogs] going to bite has something to do with dark backgrounds [in choice A].”* She further explained that she made the selection based on her impressions of what kind of dogs would bite or not, *“I would say B is the correct answer because...noticing that all the most of the dogs that say bite look like my dogs, they are small. So I’m kinda of confident about it.”* She further commented on choice D, *“D is most definitely wrong...because well, the other answers just make more sense. And also why would you ask us if there’s a flaw?”.*

Based on the feedback, we made the changes: first we removed the background of all images used in this question. Second, considering that some students may have dogs as pets and some may not, we changed the scenarios to fruits and vegetables that all students are familiar with. Finally, to prevent students from answering the question based on their prior experiences instead of the images presented in the question, we changed the question statement, emphasizing that the AI system is only trained based on the pictures provided in the question. The revised question is shown in Table 2 version #3. We conducted another round of interviews using the revised question and found no further issues in item interpretation, coherent elaboration or congruent answer choice.

Besides informing us of the revision of items, the cognitive interview findings also provided suggestions on which items to remove from the item pool. As an example, Fig. 1 shows an item assessing whether students understand

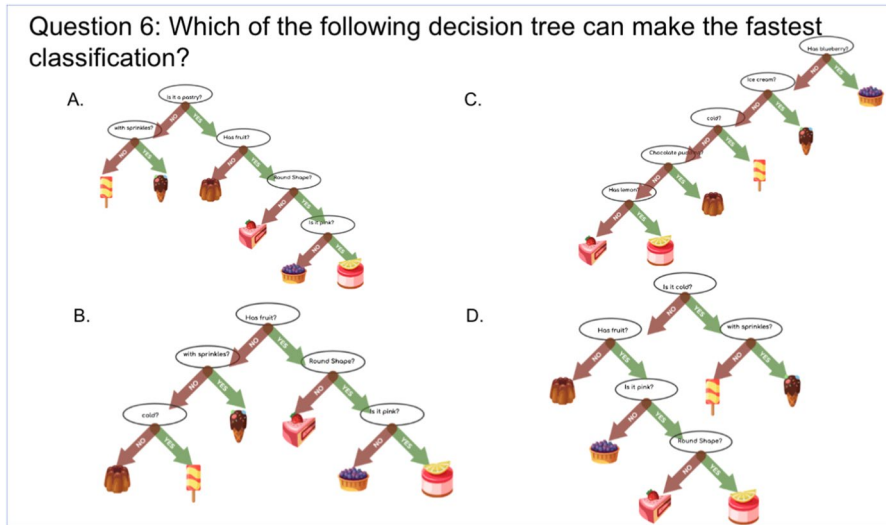


Fig. 1 An initial draft of a logic system question assessing student understanding of decision-making

decision-making in logic systems. Students were presented with four different decision trees and asked which one would classify things fastest. The interviews showed that students had different interpretations of “fastest classification” from what the test developers intended them to. For instance, student R explained that fastest classification means “*classifying things in the first step.*” She chose choice C because “*it [the decision tree] goes diagonal. You can always just go the left side... without needing to go right side. It [the decision tree] goes all the way, you will get the first one [classified] after the first sorting.*” Another student (student SY) was confused because he thought fastest classification means “*the least number of questions asked.*” His strategy was to count the number of questions asked in each decision tree, which failed because each tree included five questions. These findings suggested that students do not understand what the “fastest classification” meant. Neither do they understand a balanced decision tree would on average have the highest probability of make the fastest classification. When trying to explain this concept to middle school interviewees, we realized that most students have not learned probability yet and this concept is beyond their capability. After a few discussions with AI experts and teachers in our team, we decided to remove this question.

Quantitative Tests for Validity and Item Reduction

Based on the findings of the cognitive interviews, we revised the pool of items and ended with 30 multiple choice questions (see Appendix 2 for the whole set of 30 questions). Next, we recruited middle school teachers from three public school

districts across the United States to help collect large scale data from their students for further validation. All three districts serve a large percentage of students from historically underrepresented groups (with 59%, 85%, and 90% of students from underrepresented groups respectively). In total 981 students participated and completed the test during the spring of 2021. Among these participants, 36% identified themselves as female, 57% as male, 7% as non-binary or prefer other pronouns or refused to answer the question. The percentages of students from underrepresented groups were high: Black (35%), Hispanic or Latinx (29%), and two or more races (15%). Seven percent of the students were Asian, 8% were White, and the rest 6% refused to answer the question.

All participating students completed the AI-CI within a class period (approximately 45 min). The AI-CI was administered using Qualtrics, a safe and reliable online platform for collecting research data. Student responses to each item were coded as 0 (incorrect) or 1 (correct) and the maximum score of the 30-item test was 30. We calculated the total score of the sample: the mean score was 12.00 with a standard deviation of 3.13. An analysis of variance using gender and race/ethnicity as factors was performed to examine differences between total test score means. There was no significant interaction effect between race and gender [$F(7, 647) = 1.08, p = 0.38$], nor were significant main effects of gender [$F(2, 647) = 1.22, p = 0.30$] or race/ethnicity [$F(4, 647) = 2.23, p = 0.08$] observed. The lack of statistical significance between total test scores indicates that the performance of each group (measured by their raw score) was not significantly different.

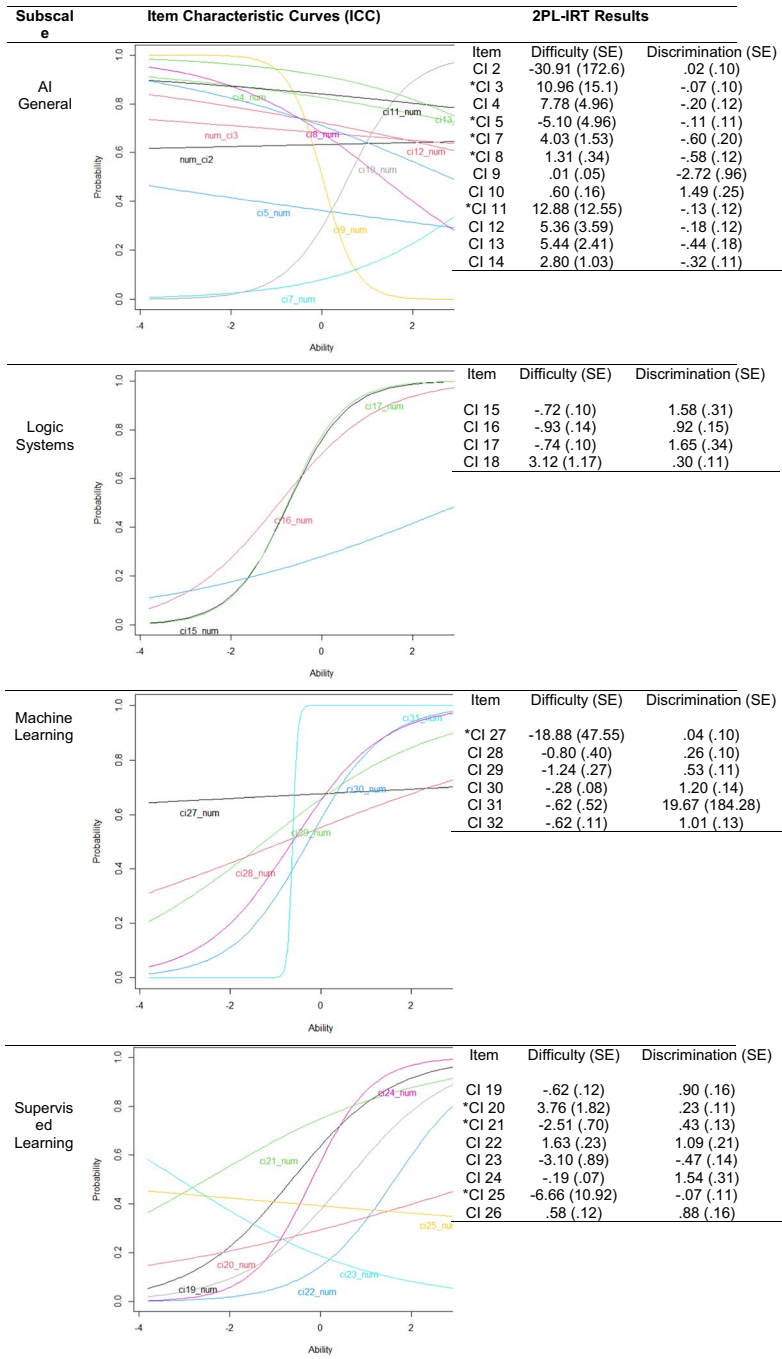
Next, characteristics of each AI-CI item were calculated using the large scale validation test data with the goal of selecting the best combination of item quality and coverage of target concepts for each topic. Test items are most often described using two parameters: difficulty (the likelihood an item will be answered correctly based on the test-taker's ability) and discrimination (the degree of how quickly the probability of a correct answer varies by latent ability). We used the Two-Parameter Logistic Item Response Theory (2PL-IRT) to calculate the parameters (Bock & Gibbons, 2021). Compared to the Classical Test Theory (CTT) approaches that typically sum the number of correct items to describe and calculate a raw student score, IRT models can better represent a test-taker's actual ability—especially when the test items have a wide range of difficulties. For example, consider two students who both took a ten-item test to measure their AI literacy. CTT fails to account for item-level difficulty and describes a student's latent ability relative to the sample's performance characteristics which is inappropriate for concept inventories. IRT models address these shortcomings and are often used to provide evidence of CI validity (Jorion et al., 2015). In particular the 2PL-IRT models used in this study characterizes a test item's difficulty and discrimination, based on the student's latent ability in the test construct's domain. The difficulty parameter is normed to zero in the item-set, meaning negative values represent easier questions and positive values represent harder questions (the more the difficulty value the harder the question). An ideal set of questions should include

both easy and difficult questions. The questions should also have high discrimination power: they have a near zero probability of being answered correctly below a latent ability threshold and then rapidly increase to probably of one above the threshold. The 2PL-IRT parameters are visualized and interpreted using item characteristic curves (ICC) which plot standardized latent ability (denoted as θ) on the x-axis and probability of responding correctly on the y-axis.

We began the quantitative validation process by creating separate 2PL-IRT models for each topic defined during the expert interview using the *ltm* package in R 4.3.2 (Rizopoulos, 2006). However, we found using the topics as subscales was problematic on the item- and factor-level. For example, some items had no discrimination across $\pm 4 \theta$ (discrimination power close to 0) meaning students with high or low latent ability performed similarly on the item, while some other items had negative discrimination meaning students with lower latent ability were more likely to answer the item correctly than high-ability students. Table 3 shows more detailed results of the ICC and 2PL-IRT model for each topic.

The detailed characteristics of the items provided suggestions on whether to include them in the final AI-CI. We prioritized removing questions that were too easy or too difficult (difficulty value too far from 0) and/or items with negative discrimination power or discrimination power close to 0, such as item CI27 (difficulty value = -18.88, discrimination value = 0.04, not discriminative), item CI11 (difficulty value = 12.88, discrimination value = -0.13), item CI3 (difficulty value = 10.96, discrimination value = -0.07), item CI5 (difficulty value = -5.10, discrimination value = -0.11), and item CI25 (difficulty value = -6.66, discrimination value = -0.07). We also compared the characteristics of items with similar difficulty and discrimination values to determine which item to be removed. For instance, item CI20 (difficulty value = 3.76, discrimination value = 0.23) and CI22 (difficulty value = 1.63, discrimination value = 1.09) were similar. After examining the question statement and item characteristics, we decided to remove CI20 because it was slightly more difficult and less discriminative than CI22. We kept item CI2 as the first question of AI-CI even though it was easy and not discriminative (difficulty value = -30.91, discrimination power = 0.02) because the literature suggests that starting an assessment from easiest questions lowers the test-takers' probability to abandon the test (Anaya et al., 2022). In total we dropped 10 items from the initial pool of questions administered to the student sample and ended with 20 items. See Appendix 1 for a list of the AI-CI items.

With the reduced set of 20 items, we tested a simpler model where each item was associated with AI literacy as one scale rather than a series of related topics associated with each other and AI literacy. The results showed satisfactory performance of the simpler model. Figure 2 is a Wright Map of the 2PL-IRT and demonstrates the performance displaying a range of student abilities in the sample on top of a distribution of item difficulty. Post hoc analysis of the model using modified parallel analysis (Dragow & Lissak, 1983) suggests that a unidimensional model performs well (see Fig. 3). Taken together, preponderance of evidence supports the validity of a twenty-item, one-dimensional "AI literacy" concept inventory.

Table 3 The Item Characteristic Curve (ICC) and 2PL-IRT model results for each planned subscale

By visual inspection, many items are reversed (i.e., students with lower ability are more likely to answer correctly than students with higher ability), are exceedingly difficult (i.e., students with ability = 4 have a

Table 3 (continued)

low probability of getting the item correct), or offer little to no discrimination in this range (i.e., the ICC has a small and near constant slope). Additionally, the magnitude of the standard errors relative to the difficulty and discrimination parameters indicate many items are a poor fit. Taken together, this evidence suggests that the subscale models, as originally designed, may not adequately reflect the knowledge structure of AI literacy for middle school students. We removed 10 items with unsatisfactory characteristics (marked with asterisk). Afterwards, we tested and refined a one-factor solution combining each subscale and found sufficient validity evidence for the reduced 20-item AI-CI

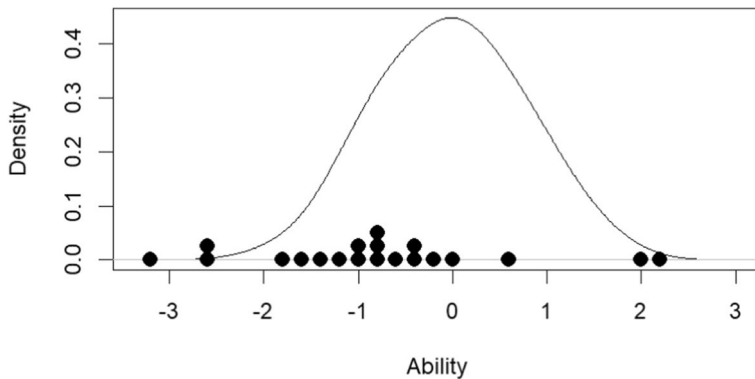


Fig. 2 The Wright Map of the modified 20-item AI-CI. *Note.* The smoothed line represents the distribution of student abilities based on the 2PL-IRT model. This is a roughly normal distribution centered around zero. Each point represents an individual item placed on its relative difficulty

Quantitative Test for Reliability (Step 6)

The sixth step focused on the pre/post-test reliability because an ideal instrument should be sensitive to a curricular intervention that teaches the target concepts. To test the pre/post-test reliability, we administered the reduced 20-item AI-CI before and after students who learned the DAILY curriculum and conducted a paired-sample t-test and Cronbach's alpha. The analysis was run in Stata 17.

In total 108 students (56% males and 43% females) consented to participate in the study. The students were middle grades students in fifth grade to eighth grade (ages 10–13). Seventh-grade students were 58% of the total participants, with fifth, sixth, and eighth graders representing 15%, 18%, and 9% of the sample respectively. A majority of participants were students from ethnic and racial groups underrepresented in STEM/CS: 33% identified as Black, 18% identified as Hispanic or Latinx, 11% identified as two races and native Americans. The remaining 38% of the sample identified themselves as Asian, White, or refused to provide their racial/ethnic information. The participants spent three months learning the DAILY curriculum in classroom. The AI-CI was administered using Qualtrics and all students completed it within 25 min.

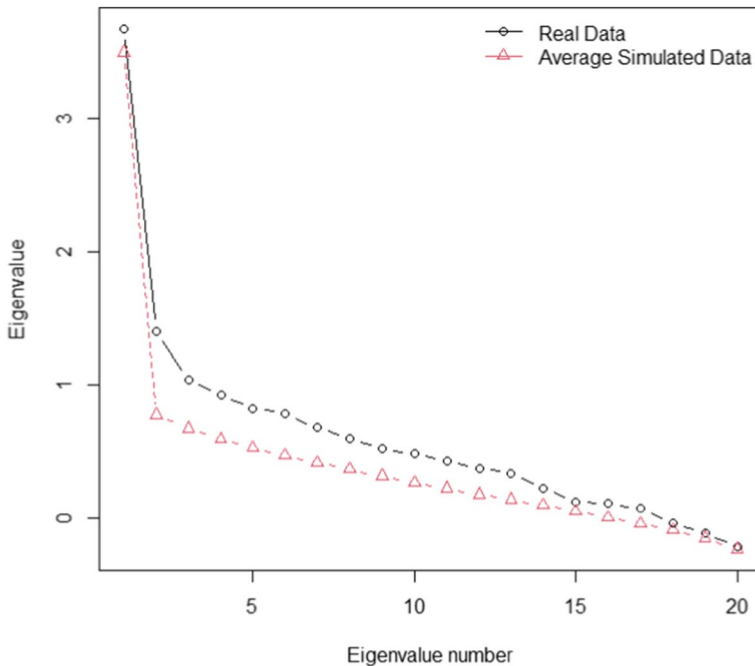


Fig. 3 Unidimensional Test using Modified Parallel Methods of the 20-item AI-CI. *Note.* The MPA was conducted using the ltm package in R. Following similar analysis patterns as a scree test in factor analysis, the “elbow” in relative eigenvalue suggests a one-factor solution in each model. Furthermore, the difference between eigenvalue first and second and second and third eigenvalues are similar by visual inspection in the real and simulated ATA

After data cleaning (i.e., removing responses that were incomplete and invalid), our study used a total of 193 data samples, in which 100 of them were post-test data, and 93 were pre-test data. Some students did not complete their pre or post-test. A calculation of the Cronbach’s alpha using pretest data with the 100 pre-test data showed a value of 0.73, greater than the 0.70 cutoff value (DeVellis, 2016). Cronbach’s alpha coefficient measures the internal consistency or reliability of a set of items, that is, how closely related a set of items are as a group. Cronbach’s alpha value is calculated by taking a score from each item and correlating it with the total score for each observation. It can help determine whether a collection of items consistently measures the same characteristic. Cronbach’s alpha quantifies the level of agreement on a standardized 0 to 1 scale. Higher values indicate that response values for each participant across a set of questions are consistent, i.e., when test-takers give a high response for one of the items, they are also likely to provide high responses for the other items. This Cronbach’s alpha value (0.73) indicated the 20-item AI-CI was reliable and the items were likely measuring the same characteristic.

Next, we conducted a paired t-test with data of 86 students who completed both the pre and post-tests. On average the students had a pretest score of

13.21 (standard deviation = 3.30) and a posttest score of 14.56 (standard deviation = 2.99). An analysis of variance using gender and race/ethnicity as factors was performed to examine differences between post-test score means. There was no significant interaction effect between race and gender [$F(4, 74) = 1.07$, $p = 0.37$], nor were significant main effects of gender [$F(2, 74) = 0.48$, $p = 0.63$] or race/ethnicity [$F(4, 74) = 1.76$, $p = 0.15$] observed. This shows that the performance of each group (measured by their post-test score) is not significantly different. The t-test showed significant increases in student AI-CI scores after experiencing the DAILY curriculum ($t(85) = 4.80$, $p < 0.0001$). Taken together, these findings suggest that these items fit together well and are sensitive to educational interventions. Students from different backgrounds (gender, race and ethnicity) do not perform significantly different on the test.

Conclusions and Future Work

This paper reports the design and validation findings of the AI-CI to measure middle school students' understanding of core AI literacy concepts. The goal was to create an AI assessment using multiple choice questions that target AI literacy concepts age appropriate for middle school students. The development of AI-CI went through a systematic procedure of six steps, among which we found it critical to conduct student interviews to ensure the questions include misconceptions as distractors. Considering their exposure to various AI technologies through social media, students have developed ideas about AI that are not documented by the literature. For instance, we found that students confused AI with automation, sensing, and algorithms. These misconceptions informed the design of the scenario-based questions in AI-CI. Once the items were developed, cognitive interviews to validate what cognitive processes and resources students went through when working on the items were valuable. In particular it was helpful to have students think aloud not only why they made the selections but also why they did not select the other choices. This revealed further insights into student misconceptions of AI and how they may misinterpret the question, which informed the refinement of the items. Afterwards we created a first draft of the AI-CI with 30 items and conducted a large scale validation study with a sample of 981 middle school students. The IRT analysis revealed detailed characteristics of the items and helped determine the reduction of items. In total we removed 10 questions that were too easy, or too difficult, or not discriminative, or had similar difficulty and discrimination values as another item. The reduced 20-item version was found to have satisfactory characteristics as one construct of AI literacy. We also examined the pre-posttest reliability of the 20-item AI-CI and found strong evidence that the instrument had acceptable internal consistency and was sensitive to educational interventions.

Our work contributes to the field of AI literacy by developing an assessment that measures the conceptual learning of age appropriate topics for middle school students using their misconceptions as distractor choices. Through the expert panels, interviews, and observations, we identified a set of core concepts that are age

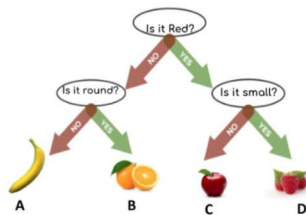
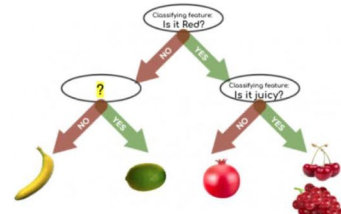
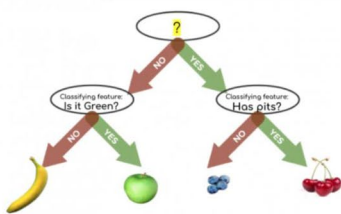
appropriate among teenagers. The interviews and observations were conducted among a group of students purposefully selected who were predominantly from groups underrepresented in STEM/CS to ensure that the analysis would capture misconceptions held by these students. The quantitative tests for validity and reliability included the analysis of variance to examine whether students from different groups responded differently on the test. Overall the AI-CI was suggested to be valid, reliable and sensitive to learning interventions.

Our work also contributes to the field of concept inventory development in that we used the validation to help inform the reduction of items so that middle school students can complete within a class period. Item reduction is critical when developing assessments for teenagers because their attention span is shorter than adults, particularly those who are heavy exposed to social media nowadays (Sriram, 2023). Different from concept inventories developed for undergraduates or other adults, an assessment with an excessive number of items can lead to fatigue among adolescents (Porter et al., 2004) and declines in completion rates. This is particularly true for youth from underrepresented groups because they often have low self-esteem in terms of their academic capabilities. Therefore, concept inventories developed for young learners typically should take less than 30 min for them to complete. For example, the Middle Grades Computer Science Concept Inventory (MG-CSCI) Assessment (Rachmatullah et al., 2020) included 24 items and the Moon Phases Concept Inventory for Middle School (MPCI-MS) include 19 items (Chastenay & Riopel, 2020). Using the IRT results we managed to remove 10 questions and the 20-item version of AI-CI can be easily completed within 25 min.

As our next step, we plan to further validate the AI-CI items by conducting additional structural analysis to investigate why a unidimensional model of AI literacy presents stronger evidence for validity than using the topics. Prior research has suggested that such item-level analysis of concept inventories can be used to improve test-writing to ensure distractors function correctly (i.e., are more like to selected by students with lower AI literacy than higher AI literacy) and inform future curriculum and instructional design (Jorion et al., 2015). Further given our target audience are middle school students from underrepresented groups, we will conduct closer examinations of the performance of students from different racial and ethnicity groups from pretest to posttest to identify if and to what extent certain misconceptions were prominent and/or stubborn among certain group of students. This would reveal more information about student learning and inform the design of curricular approaches for students of certain groups. In addition, although our expert panels expressed concerns over the age appropriateness of teaching neural networks and GANs, we wonder if middle school students are able to develop fundamental understandings of these two concepts and how assessments like AI-CI can capture student learning of these two topics. Therefore, as a next step we will engage in a similar process as described earlier in this paper to develop items aiming to capture student understanding of neural networks and GANs. Finally, concept inventory as an assessment tool needs to be designed to be curricular independent. Conducting another pre and post-test reliability study with another group of middle school youth who learn with an AI literacy curriculum other than DAILY can reveal more information about the assessment and provide more evidence to determine the reliability of AI-CI.

Appendix 1: The list of the 20-item AI-CI

Item code	Question									
CI2 & CI4	Review the following technologies and indicate whether or not they involve artificial intelligence (AI) <table><tr><td></td><td>AI</td><td>Not AI</td></tr><tr><td>A software that can classify different types of hats based on the type</td><td>☐</td><td>☐</td></tr><tr><td>Facial recognition software that can tell whether you or your friends are in the picture</td><td>☐</td><td>☐</td></tr></table>		AI	Not AI	A software that can classify different types of hats based on the type	☐	☐	Facial recognition software that can tell whether you or your friends are in the picture	☐	☐
	AI	Not AI								
A software that can classify different types of hats based on the type	☐	☐								
Facial recognition software that can tell whether you or your friends are in the picture	☐	☐								
CI9	An automatic door opens when someone steps onto the mat in front of it. This mat includes a pressure sensor that sends a signal to the door to open. Does this automatic door use a dataset? ☐ Yes ☐ No									
CI12	A phone app keeps a record of the types of games you played before. When a new game is released that you might like, it will tell you about it. Does this app make a prediction? ☐ Yes ☐ No									
CI13 & CI14	A school stores all of its students' contact information in a computer program. This student data includes their email address, phone number, and address. This school uses a software to help them search for information about certain students. For example, the school could identify all of the 4th grade students. <i>Answer the following questions based on the technology described above.</i> <table><tr><td></td><td>Yes</td><td>No</td></tr><tr><td>Does this computer program use a dataset?</td><td>☐</td><td>☐</td></tr><tr><td>Does this computer program make a prediction?</td><td>☐</td><td>☐</td></tr></table>		Yes	No	Does this computer program use a dataset?	☐	☐	Does this computer program make a prediction?	☐	☐
	Yes	No								
Does this computer program use a dataset?	☐	☐								
Does this computer program make a prediction?	☐	☐								

CI15	Computers make use of decision trees to classify data. Below is an image of a decision tree.  Imagine you categorize a blueberry using this decision tree.  In which category will it end up? ☐ A ☐ B ☐ C ☐ D
CI16	Below is an image of a decision tree.  Pick which question was most likely used at the "?" oval to classify this set of fruit. ☐ Is it sour? ☐ Is it small? ☐ Is it blue? ☐ Is it long?
CI17	Below is an image of a decision tree.  Pick which question was most likely used at the "?" oval to classify the set of fruit. ☐ Is it small? ☐ Is it red? ☐ Is it large? ☐ Is it blue?

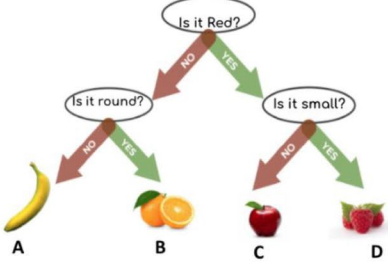
CI18	Humans make decision trees to classify data. Below is a diagram of the steps a person takes during this process. The backwards arrow from Step 3 to Step 2 indicates that steps 2 and 3 happen repeatedly.  In step 1, the person picks a dataset to use. What should the person do in steps 2 and 3? ☐ Split the set based on the question; then pick a feature of the data to ask about. ☐ Pick a feature of the data to ask about; then split the set based on the question. ☐ Pick a feature of the data to ask about; then decide what the leaves of the tree would be. ☐ Decide what the leaves of the tree would be; then pick a feature of the data to ask about.
CI28	A doctor sends x-ray scans out to an AI to determine whether or not you have broken a bone. Is this an example of an AI classifying or a generating? ☐ Classifying ☐ Generating
CI29	An app converts the style of your face to a Van Gogh painting. Is this an example of an AI classifying or a generating? ☐ Classifying ☐ Generating
CI30	A machine is given a dataset of thousands of audio clips (containing sounds only) and generates a new audio clip similar to the dataset. ☐ Unsupervised Learning ☐ Supervised Learning
CI31	A machine is given a dataset of images of flowers with the identification of whether the image is a of a "tulip" or "not tulip" and is trained to identify when an image contains a tulip. ☐ Unsupervised Learning ☐ Supervised Learning

CI32	A machine is given a dataset of images of dogs (without naming them) and can group different dogs by their types. ☐ Unsupervised Learning ☐ Supervised Learning
CI19	An AI technology has been trained to classify images only using the following data: Banana  Broccoli  Banana  Broccoli  Which of the following would you expect the technology to predict? ☐ All crunchy things are broccoli ☐ All yellow things are bananas ☐ All things in multiples are broccoli ☐ All things in multiples are bananas
CI22	An AI technology has been trained to classify images only using the following data: Broccoli  Banana  Given the two images of bunches of broccoli and bananas below, which of the two will the AI technology be more likely to recognize correctly?   ☐ Broccoli ☐ It will be the same for broccoli and bananas ☐ Bananas
CI23	Samantha uses labeled pictures to train an AI technology to tell if a picture has her in it or not. During the training phase, if the technology identifies incorrectly, what does it do to improve the system? ☐ Strengthen or weaken connections in the network ☐ Ask the algorithm to remember the picture that AI technology identifies wrong ☐ Delete the picture ☐ Change the label of the picture so that the AI technology won't identify it incorrectly

CI24	An AI technology can recognize men's shoes accurately but not women's shoes. Which of the following could be the cause? ☐ The dataset it uses to learn to identify shoes only includes men's shoes ☐ Some brand labels don't make women's shoes ☐ Women's shoes are smaller than men's shoes, making them difficult to identify ☐ The dataset it uses to learn to identify shoes only includes women's shoes
CI26	What is the purpose of a label on an image in Supervised Learning? ☐ The label is used to check if the prediction made was correct ☐ The label is used to see if the image contains words ☐ The label is used to alert the human supervisor to classify a model ☐ The label is not useful in supervised learning

Appendix 2: The list of the initial draft of 30 items of AI-CI

Item code	Question																		
CI2-5	Review the following technologies and indicate whether or not they involve artificial intelligence (AI) <table><thead><tr><th></th><th>AI</th><th>Not AI</th></tr></thead><tbody><tr><td>A software that can classify different types of hats</td><td>☐</td><td>☐</td></tr><tr><td>A graphics app that can generate new paintings by combining features from other paintings</td><td>☐</td><td>☐</td></tr><tr><td>Facial recognition software that can tell whether you or your friends are in the picture</td><td>☐</td><td>☐</td></tr><tr><td>A smartphone app that reminds people to turn off the lights in their house if the time is between 9am to 3pm</td><td>☐</td><td>☐</td></tr></tbody></table>		AI	Not AI	A software that can classify different types of hats	☐	☐	A graphics app that can generate new paintings by combining features from other paintings	☐	☐	Facial recognition software that can tell whether you or your friends are in the picture	☐	☐	A smartphone app that reminds people to turn off the lights in their house if the time is between 9am to 3pm	☐	☐			
	AI	Not AI																	
A software that can classify different types of hats	☐	☐																	
A graphics app that can generate new paintings by combining features from other paintings	☐	☐																	
Facial recognition software that can tell whether you or your friends are in the picture	☐	☐																	
A smartphone app that reminds people to turn off the lights in their house if the time is between 9am to 3pm	☐	☐																	
CI 7-10	There's an app that tracks how much time you spend on social media per day. If you spend more than an hour on social media, the app will ban you from the site for the rest of the day. <i>Answer the following questions based on the technology described above.</i> <table><thead><tr><th></th><th>Yes</th><th>No</th></tr></thead><tbody><tr><td>Does this app use a dataset?</td><td>☐</td><td>☐</td></tr><tr><td>Does this app make a prediction?</td><td>☐</td><td>☐</td></tr></tbody></table> An automatic door opens when someone steps onto the mat in front of it. This mat includes a pressure sensor that sends a signal to the door to open. <i>Answer the following questions based on the technology described above.</i> <table><thead><tr><th></th><th>Yes</th><th>No</th></tr></thead><tbody><tr><td>Does this automatic door use a dataset?</td><td>☐</td><td>☐</td></tr><tr><td>Does this automatic door make a prediction?</td><td>☐</td><td>☐</td></tr></tbody></table>		Yes	No	Does this app use a dataset?	☐	☐	Does this app make a prediction?	☐	☐		Yes	No	Does this automatic door use a dataset?	☐	☐	Does this automatic door make a prediction?	☐	☐
	Yes	No																	
Does this app use a dataset?	☐	☐																	
Does this app make a prediction?	☐	☐																	
	Yes	No																	
Does this automatic door use a dataset?	☐	☐																	
Does this automatic door make a prediction?	☐	☐																	

CI 11-12	A phone app keeps a record of the types of games you played before. When a new game is released that you might like, it will tell you about it. <i>Answer the following questions based on the technology described above.</i> <table><thead><tr><th></th><th>Yes</th><th>No</th></tr></thead><tbody><tr><td>Does this app use a dataset?</td><td>☐</td><td>☐</td></tr><tr><td>Does this app make a prediction?</td><td>☐</td><td>☐</td></tr></tbody></table>		Yes	No	Does this app use a dataset?	☐	☐	Does this app make a prediction?	☐	☐
	Yes	No								
Does this app use a dataset?	☐	☐								
Does this app make a prediction?	☐	☐								
CI 13-14	A school stores all of its students' contact information in a computer program. This student data includes their email address, phone number, and address. This school uses a software to help them search for information about certain students. For example, the school could identify all of the 4th grade students. <i>Answer the following questions based on the technology described above.</i> <table><thead><tr><th></th><th>Yes</th><th>No</th></tr></thead><tbody><tr><td>Does this computer program use a dataset?</td><td>☐</td><td>☐</td></tr><tr><td>Does this computer program make a prediction?</td><td>☐</td><td>☐</td></tr></tbody></table>		Yes	No	Does this computer program use a dataset?	☐	☐	Does this computer program make a prediction?	☐	☐
	Yes	No								
Does this computer program use a dataset?	☐	☐								
Does this computer program make a prediction?	☐	☐								
CI15	Computers make use of decision trees to classify data. Below is an image of a decision tree.  Imagine you categorize a blueberry using this decision tree.  In which category will it end up? ☐ A ☐ B ☐ C ☐ D									

CI16	Below is an image of a decision tree. <pre> graph TD Q1((?)) -- NO --> B[Banana] Q1 -- YES --> Q2((Classifying feature: Is it Red?)) Q2 -- NO --> P[Pomegranate] Q2 -- YES --> G[Grapes] </pre> Pick which question was most likely used at the "?" oval to classify this set of fruit. ☐ Is it sour? ☐ Is it small? ☐ Is it blue? ☐ Is it long?
CI17	Below is an image of a decision tree. <pre> graph TD Q1((?)) -- NO --> B[Banana] Q1 -- YES --> Q2((Classifying feature: Is it Green?)) Q2 -- NO --> BB[Blueberries] Q2 -- YES --> C[Cherries] </pre> Pick which question was most likely used at the "?" oval to classify the set of fruit. ☐ Is it small? ☐ Is it red? ☐ Is it large? ☐ Is it blue?

CI18	Humans make decision trees to classify data. Below is a diagram of the steps a person takes during this process. The backwards arrow from Step 3 to Step 2 indicates that steps 2 and 3 happen repeatedly.  In step 1, the person picks a dataset to use. What should the person do in steps 2 and 3? ☐ Split the set based on the question; then pick a feature of the data to ask about. ☐ Pick a feature of the data to ask about; then split the set based on the question. ☐ Pick a feature of the data to ask about; then decide what the leaves of the tree would be. ☐ Decide what the leaves of the tree would be; then pick a feature of the data to ask about.
CI27-28	A social media site just released a feature that allows you to upload a photo of yourself, and the system will recognize your face and tag your name. Is this an example of an AI classifying or a generating? ☐ Classifying ☐ Generating A doctor sends x-ray scans out to an AI to determine whether or not you have broken a bone. Is this an example of an AI classifying or a generating? ☐ Classifying ☐ Generating
CI29	An app converts the style of your face to a Van Gogh painting. Is this an example of an AI classifying or a generating? ☐ Classifying ☐ Generating
CI30	A machine is given a dataset of thousands of audio clips (containing sounds only) and generates a new audio clip similar to the dataset. ☐ Unsupervised Learning ☐ Supervised Learning

CI31	A machine is given a dataset of images of flowers with the identification of whether the image is a of a "tulip" or "not tulip" and is trained to identify when an image contains a tulip. ☐ Unsupervised Learning ☐ Supervised Learning
CI32	A machine is given a dataset of images of dogs (without naming them) and can group different dogs by their types. ☐ Unsupervised Learning ☐ Supervised Learning
CI19	An AI technology has been trained to classify images only using the following data: Which of the following would you expect the technology to predict? ☐ All crunchy things are broccoli ☐ All yellow things are bananas ☐ All things in multiples are broccoli ☐ All things in multiples are bananas
CI20	An AI technology has been trained to classify images only using the following data: Based on the above dataset, how would it classify the image below? ☐ Banana ☐ Broccoli

CI21	An AI technology has been trained to classify images only using the following data: Broccoli  Banana 
	Based on the above dataset, how would it classify the following image below? 
	☐ Broccoli ☐ Banana
CI22	An AI technology has been trained to classify images only using the following data: Broccoli  Banana 
	Given the two images of bunches of broccoli and bananas below, which of the two will the AI technology be more likely to recognize correctly? 
	☐ Broccoli ☐ It will be the same for broccoli and bananas ☐ Bananas

CI23	Samantha uses labeled pictures to train an AI technology to tell if a picture has her in it or not. During the training phase, if the technology identifies incorrectly, what does it do to improve the system? ☐ Strengthen or weaken connections in the network ☐ Ask the algorithm to remember the picture that AI technology identifies wrong ☐ Delete the picture ☐ Change the label of the picture so that the AI technology won't identify it incorrectly
CI24	An AI technology can recognize men's shoes accurately but not women's shoes. Which of the following could be the cause? ☐ The dataset it uses to learn to identify shoes only includes men's shoes ☐ Some brand labels don't make women's shoes ☐ Women's shoes are smaller than men's shoes, making them difficult to identify ☐ The dataset it uses to learn to identify shoes only includes women's shoes
CI25	Use the definitions below to answer the next question: <u>Supervised Learning</u>: when computers learn categories or tasks by finding patterns in labeled data. <u>Training phase</u>: when an AI system is built using input data. <u>Testing phase</u>: when an AI system is tested to see how good it is at making predictions. In supervised learning, the training phase and the testing phase differ in... ☐ Where each phase takes places ☐ The dataset used in each phase ☐ Whether or not the data is labeled ☐ How the prediction is scored
CI26	What is the purpose of a label on an image in Supervised Learning? ☐ The label is used to check if the prediction made was correct ☐ The label is used to see if the image contains words ☐ The label is used to alert the human supervisor to classify a model ☐ The label is not useful in supervised learning

Acknowledgements This work was funded by the National Science Foundation award #2022502 and #2048746. We are grateful to the teachers and students involved in this study. We thank Safinah Ali, Daniella DiPaola, Nathaniel Brown, and Olivia Szendey for assisting us with item development, data collection and analysis.

Declarations

Ethics Approval The work presented in this paper was approved by MIT's Internal review Board, Committee on the Use of Humans as Experimental Subjects (COUHES) under approval no. 1912000072 and Boston College's Internal Review Board, under approval no. 20.207.01e.

Informed Consent to Participate and Publish All participants signed an assent form to participate in the study, which indicated that we will be collecting their audio, video, and activity data. Since participants were minors, all participants' parents provided written consent prior to enrollment in the study. Students were told that they can withdraw from the study at any point, and they may choose to turn off their audio and video streaming if they preferred.

Conflict of Interest The authors have no competing interests to declare that are relevant to the content of this article. The authors have no relevant financial or non-financial interests to disclose.

References

- Adams, W. K., & Wieman, C. E. (2011). Development and validation of instruments to measure learning of expert-like thinking. *International Journal of Science Education*, 33(9), 1289–1312.
- Ali, S., DiPaola, D., Lee, I., Hong, J., & Breazeal, C. (2021). Exploring generative models with middle school students. *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3411764.3445226>
- Ali, S., Payne, B. H., Williams, R., Park, H. W., & Breazeal, C. (2019). Constructionism, ethics, and creativity: Developing primary and middle school artificial intelligence education. *International Workshop on Education in Artificial Intelligence K-12 (EDUAI'19)*, 1–4.
- Almstrum, V. L., Henderson, P. B., Harvey, V., Heeren, C., Marion, W., Riedesel, C., Soh, L.-K., & Tew, A. E. (2006). Concept inventories in computer science for the topic discrete mathematics. *Working Group Reports on ITiCSE on Innovation and Technology in Computer Science Education*, 132–145. <https://doi.org/10.1145/1189215.1189182>
- Anaya, L., Iriberry, N., Rey-Biel, P., & Zamarro, G. (2022). Understanding performance in test taking: The role of question difficulty order. *Economics of Education Review*, 90, 102293. <https://doi.org/10.1016/j.econedurev.2022.102293>
- Ascalon, M. E., Meyers, L. S., Davis, B. W., & Smits, N. (2007). Distractor similarity and item-stem structure: Effects on item difficulty. *Applied Measurement in Education*, 20(2), 153–170.
- Bock, R. D., & Gibbons, R. D. (2021). *Item response theory*. Wiley.
- Brennan, K., & Resnick, M. (2012). New frameworks for studying and assessing the development of computational thinking. *Proceedings of the 2012 Annual Meeting of the American Educational Research Association, Vancouver, Canada*, 1–25.
- Briggs, D. C., Alonzo, A. C., Schwab, C., & Wilson, M. (2006). Diagnostic assessment with ordered multiple-choice items. *Educational Assessment*, 11(1), 33–63.
- Caceffo, R., Wolfman, S., Booth, K. S., & Azevedo, R. (2016). Developing a computer science concept inventory for introductory programming. *Proceedings of the 47th ACM Technical Symposium on Computing Science Education*, 364–369. <https://doi.org/10.1145/2839509.2844559>
- Cantú-Ortiz, F. J., Galeano Sánchez, N., Garrido, L., Terashima-Marin, H., & Brena, R. F. (2020). An artificial intelligence educational strategy for the digital transformation. *International Journal on Interactive Design and Manufacturing (IJIDeM)*, 14, 1195–1209.
- Celik, I. (2023). Exploring the determinants of Artificial Intelligence (AI) Literacy: Digital divide, computational thinking. *Cognitive Absorption. Telematics and Informatics*, 83, 102026. <https://doi.org/10.1016/j.tele.2023.102026>

- Chastenay, P., & Riopel, M. (2020). Development and validation of the moon phases concept inventory for middle school. *Physical Review Physics Education Research*, 16(2), 020107.
- Howitt, S., Anderson, T., Costa, M., Hamilton, S., & Wright, T. (2008). A concept inventory for molecular life sciences: How will it help your teaching practice? *Australian Biochemist*, 39(3), 14–17.
- Cowie, B., & Bell, B. (1999). A model of formative assessment in science education. *Assessment in Education: Principles, Policy & Practice*, 6(1), 101–116.
- Crouch, C. H., & Mazur, E. (2001). Peer instruction: Ten years of experience and results. *American Journal of Physics*, 69(9), 970–977.
- DeVellis, R. F. (2016). *Scale development: Theory and applications* (Vol. 26). Sage Publications.
- Drasgow, F., & Lissak, R. I. (1983). Modified parallel analysis: A procedure for examining the latent dimensionality of dichotomously scored item responses. *Journal of Applied Psychology*, 68(3), 363.
- Druga, S., Williams, R., Breazeal, C., & Resnick, M. (2017). “Hey Google is it OK if I eat you?”: Initial explorations in child-agent interaction. *Proceedings of the 2017 Conference on Interaction Design and Children*, 595–600. <https://doi.org/10.1145/3078072.3084330>
- Eguchi, A., Okada, H., & Muto, Y. (2021). Contextualizing AI education for K-12 students to enhance their learning of AI literacy through culturally responsive approaches. *KI-Künstliche Intelligenz*, 35(2), 153–161.
- Fast, E., & Horvitz, E. (2017). Long-term trends in the public perception of artificial intelligence. *Proceedings of the AAI Conference on Artificial Intelligence*, 31(1). <https://doi.org/10.1609/aaai.v31i1.10635>
- Garvin-Doxas, K., & Klymkowsky, M. W. (2008). Understanding randomness and its impact on student learning: Lessons learned from building the Biology Concept Inventory (BCI). *CBE—Life Sciences Education*, 7(2), 227–233.
- Goldman, K., Gross, P., Heeren, C., Herman, G. L., Kaczmarczyk, L., Loui, M. C., & Zilles, C. (2010). Setting the scope of concept inventories for introductory computing subjects. *ACM Transactions on Computing Education (TOCE)*, 10(2), 1–29.
- Guerreiro-Santalla, S., Bellas, F., & Duro, R. J. (2020). Artificial intelligence in pre-university education: What and how to teach. *Multidisciplinary Digital Publishing Institute Proceedings*, 54. <https://doi.org/10.3390/proceedings2020054048>
- Hestenes, D., Wells, M., & Swackhamer, G. (1992). Force concept inventory. *The Physics Teacher*, 30(3), 141–158.
- Jorion, N., Gane, B. D., James, K., Schroeder, L., DiBello, L. V., & Pellegrino, J. W. (2015). An analytic framework for evaluating the validity of concept inventory claims. *Journal of Engineering Education*, 104(4), 454–496.
- Kandlhofer, M., Steinbauer, G., Hirschmugl-Gaisch, S., & Huber, P. (2016). Artificial intelligence and computer science in education: From kindergarten to university. *IEEE Frontiers in Education Conference (FIE)*, 2016, 1–9. <https://doi.org/10.1109/FIE.2016.7757570>
- Karabenick, S. A., Woolley, M. E., Friedel, J. M., Ammon, B. V., Blazevski, J., Bonney, C. R., Groot, E. D., Gilbert, M. C., Musu, L., & Kempler, T. M. (2007). Cognitive processing of self-report items in educational research: Do they think what we mean? *Educational Psychologist*, 42(3), 139–151.
- Krause, S., Birk, J., Bauer, R., Jenkins, B., & Pavelich, M. J. (2004). Development, testing, and application of a chemistry concept inventory. *34th Annual Frontiers in Education, 2004. FIE 2004.*, T1G-1. <https://doi.org/10.1109/FIE.2004.1408473>
- Kreinsen, M., & Schulz, S. (2021). Students’ conceptions of artificial intelligence. *Proceedings of the 16th Workshop in Primary and Secondary Computing Education*. <https://doi.org/10.1145/3481312.3481328>
- Lane, D. (2021). *Machine learning for kids: A project-based introduction to artificial intelligence*. No Starch Press.
- Libarkin, J. C., Anderson, S. W., Deeds, D., & Callen, B. (2006). Development of the geoscience concept inventory. *Proceedings of the National STEM Assessment Conference*, 148–158.
- Lin, C. H., Wu, L., Wang, W. C., Wu, P. L., & Cheng, S. Y. (2020). *Development and validation of an instrument for AI-Literacy*. the 3rd Eurasian Conference on Educational Innovation (ECEI 2020), Hanoi, Vietnam.
- Long, D., & Magerko, B. (2020). What is AI Literacy? Competencies and design considerations. *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, 1–16. <https://doi.org/10.1145/3313831.3376727>
- Mazur, E. (2009). Farewell, lecture? *Science*, 323(5910), 50–51.

- Messick, S. (1995). Validity of psychological assessment: Validation of inferences from persons' responses and performances as scientific inquiry into score meaning. *American Psychologist*, 50(9), 741.
- Morris, G. A., Branum-Martin, L., Harshman, N., Baker, S. D., Mazur, E., Dutta, S., Mzoughi, T., & McCauley, V. (2006). Testing the test: Item response curves and test quality. *American Journal of Physics*, 74(5), 449–453. <https://doi.org/10.1119/1.2174053>
- National Science Board. (2020). *Production and trade of knowledge—And technology-intensive industries* (Science & Engineering Indicators). Retrieved April 6, 2024, from <https://nces.nsf.gov/pubs/nsb20205>
- Ng, D. T. K., Leung, J. K. L., Chu, S. K. W., & Qiao, M. S. (2021). Conceptualizing AI literacy: An exploratory review. *Computers and Education: Artificial Intelligence*, 2, 100041.
- O'Connor, B. P. (1995). Identity development and perceived parental behavior as sources of adolescent egocentrism. *Journal of Youth and Adolescence*, 24(2), 205–227.
- Ottenbreit-Leftwich, A., Glazewski, K., Jeon, M., Hmelo-Silver, C., Mott, B., Lee, S., & Lester, J. (2021). How do elementary students conceptualize artificial intelligence? *Proceedings of the 52nd ACM Technical Symposium on Computer Science Education*, 1261. <https://doi.org/10.1145/3408877.3439642>
- Porter, S. R., Whitcomb, M. E., & Weitzer, W. H. (2004). Multiple surveys of students and survey fatigue. *New Directions for Institutional Research*, 2004(121), 63–73.
- Rachmatullah, A., Akram, B., Boulden, D., Mott, B., Boyer, K., Lester, J., & Wiebe, E. (2020). Development and validation of the middle grades computer science concept inventory (MG-CSCI) assessment. *EURASIA Journal of Mathematics, Science and Technology Education*, 16(5), em1841.
- Rizopoulos, D. (2006). ltm: An R package for latent variable modeling and item response analysis. *Journal of Statistical Software*, 17, 1–25.
- Robinson, O. C. (2014). Sampling in interview-based qualitative research: A theoretical and practical guide. *Qualitative Research in Psychology*, 11(1), 25–41.
- Sadler, P. M., Coyle, H., Miller, J. L., Cook-Smith, N., Dussault, M., & Gould, R. R. (2010). The astronomy and space science concept inventory: Development and validation of assessment instruments aligned with the k–12 national science standards. *Astronomy Education Review*, 8(1), 010111.
- Savinainen, A., & Scott, P. (2002). Using the Force Concept Inventory to monitor student learning and to plan teaching. *Physics Education*, 37(1), 53.
- Sriram, K. (2023). To what extent does social media usage impact the ability to delay gratification and attention span of teenagers in Mumbai? *British Journal of Multidisciplinary and Advanced Studies*, 4(6), 71–86.
- Taras, M. (2005). Summative and formative assessment—some theoretical reflections. *British Journal of Educational Studies*, 53, 466–478.
- Taylor, C., Clancy, M., Webb, K. C., Zingaro, D., Lee, C., & Porter, L. (2020). The practical details of building a cs concept inventory. *Proceedings of the 51st ACM Technical Symposium on Computer Science Education*, 372–378. <https://doi.org/10.1145/3328778.3366903>
- Taylor, C., Zingaro, D., Porter, L., Webb, K. C., Lee, C. B., & Clancy, M. (2014). Computer science concept inventories: Past and future. *Computer Science Education*, 24(4), Article 4.
- Touretzky, D., Gardner-McCune, C., Martin, F., & Seehorn, D. (2019). Envisioning AI for K-12: What should every child know about AI? *Proceedings of the AAAI Conference on Artificial Intelligence*, 33 (1), 9795–9799. <https://doi.org/10.1609/aaai.v33i01.33019795>
- Turkle, S., Breazeal, C., Dasté, O., & Scassellati, B. (2006). Encounters with kismet and cog: Children respond to relational artifacts. In L. Humphreys & P. Messaris (Eds.), *Digital media: Transformations in human communication* (pp. 313–330). Peter Lang.
- Williams, R., Ali, S., Devasia, N., DiPaola, D., Hong, J., Kaputsos, S. P., Jordan, B., & Breazeal, C. (2023). AI+ ethics curricula for middle school youth: Lessons learned from three project-based curricula. *International Journal of Artificial Intelligence in Education*, 33, 325–383. <https://doi.org/10.1007/s40593-022-00298-y>
- Williams, R., & Breazeal, C. (2020). How to train your robot: A middle school AI and ethics curriculum. International Workshop on Education in Artificial Intelligence K-12 (EDUAI '20).
- Williams, R., Park, H. W., & Breazeal, C. (2019). A is for Artificial Intelligence: The impact of artificial intelligence activities on young children's perceptions of robots. *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 1–11. <https://doi.org/10.1145/3290605.3300677>

- Woolley, M. E., Bowen, G. L., & Bowen, N. K. (2006). The development and evaluation of procedures to assess child self-report item validity educational and psychological measurement. *Educational and Psychological Measurement*, 66(4), 687–700.
- Yu, S., & Wolters, C. A. (2002). Issues in the assessment of motivation in students from ethnic minority populations. *Advances in Motivation and Achievement*, 12, 349–380.
- Zhang, H., Lee, I., Ali, S., DiPaola, D., Cheng, Y., & Breazeal, C. (2023). Integrating ethics and career futures with technical learning to promote AI literacy for middle school students: An exploratory study. *International Journal of Artificial Intelligence in Education*, 33(2), 290–324.
- Zhou, X., Van Brummelen, J., & Lin, P. (2020). Designing AI learning experiences for K-12: Emerging works, future opportunities and a design framework. *arXiv Preprint arXiv:2009.10228*.
- Zimmerman, M. (2018). *Teaching AI: Exploring new frontiers for learning*. International Society for Technology in Education.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.